2.2 Does contract testing replace other types of tests?

You may wonder whether having contract testing in place eliminates other types of tests. Hopefully, by now, you get the idea that this isn't the case, especially with the introduction of the revised test automation pyramid. Contract testing doesn't replace these tests but rather complements them.

To prove this point, let's consider a simple scenario. Imagine that your team is working on a new login feature. During the development process, you and your team collectively identify the following 10 potential failures for this feature:

1. The username is null.
2. The username is empty.
3. The password is null.
4. The password is empty.
5. The username exceeds the limit of 20 alphanumeric characters.
6. An unauthorized user tries to log in.
7. A user tries to log in but with an incorrect password.
8. A user tries to log in but with an incorrect username.
9. The login service is responding slow due to a bottleneck of incoming requests.
10. A user clears their browser cookies and their session hasn't been cleared.

If we reference the revised test automation pyramid in figure 2.6, let's assume that scenarios 1 to 5 can be captured at the unit level, while scenarios 6 to 10 require some form of verification involving other systems. From scenarios 6 to 10, scenarios 6 to 8 can be identified through contract testing, scenario 9 through integration testing, and scenario 10 through E2E testing. Now, without contract testing in place, as illustrated on the left side of figure 2.7, scenarios 6 to 9 are potentially caught on the integration level. This complicates the debugging process and introduces a slower feedback loop because integration tests are positioned higher in the